

ASSESSMENT 8.5

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BATCH 45

Task Description #1 (Username Validator – Apply AI in Authentication Context)

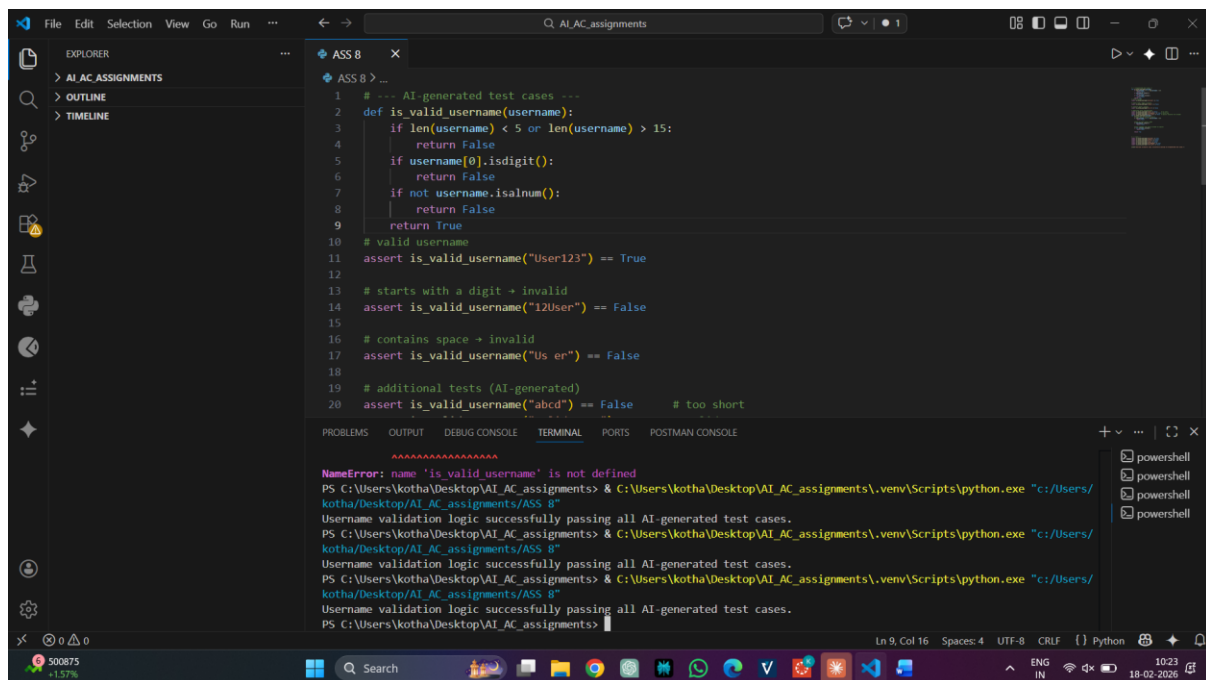
- Task: Use AI to generate at least 3 assert test cases for a function `is_valid_username(username)` and then implement the function using Test-Driven Development principles.
- Requirements:
 - o Username length must be between 5 and 15 characters.
 - o Must contain only alphabets and digits.
 - o Must not start with a digit.
 - o No spaces allowed.

Example Assert Test Cases:

```
assert is_valid_username("User123") == True
```

```
assert is_valid_username("12User") == False
```

```
assert is_valid_username("Us er") == False
```



```
1 # --- AI-generated test cases ---
2 def is_valid_username(username):
3     if len(username) < 5 or len(username) > 15:
4         return False
5     if username[0].isdigit():
6         return False
7     if not username.isalnum():
8         return False
9     return True
10 # valid username
11 assert is_valid_username("User123") == True
12
13 # starts with a digit + invalid
14 assert is_valid_username("12User") == False
15
16 # contains space + invalid
17 assert is_valid_username("Us er") == False
18
19 # additional tests (AI-generated)
20 assert is_valid_username("abcd") == False # too short
```

```
*****
NameError: name 'is_valid_username' is not defined
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function `classify_value(x)` and implement it using conditional logic and loops.

- Requirements:

- o If input is an integer, classify as "Even" or "Odd".

- o If input is 0, return "Zero".

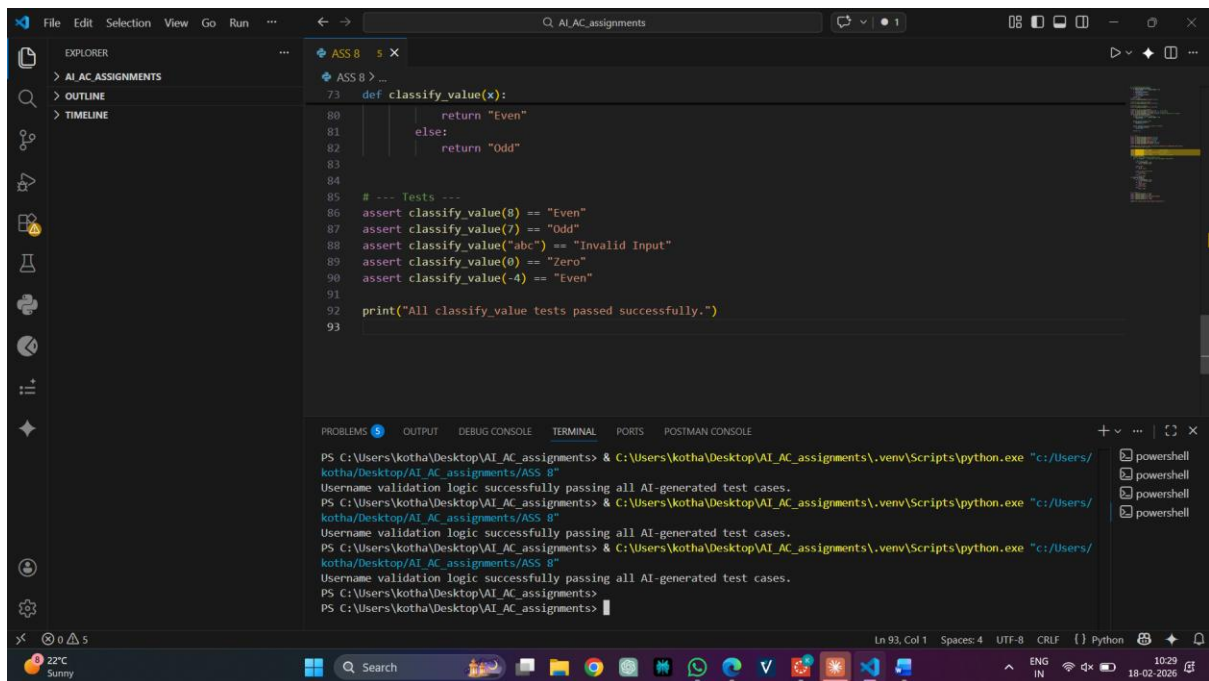
- o If input is non-numeric, return "Invalid Input".

Example Assert Test Cases:

```
assert classify_value(8) == "Even"
```

```
assert classify_value(7) == "Odd"
```

```
assert classify_value("abc") == "Invalid Input"
```



```
73 def classify_value(x):
80     return "Even"
81     else:
82         return "Odd"
83
84
85 # --- Tests ---
86 assert classify_value(8) == "Even"
87 assert classify_value(7) == "Odd"
88 assert classify_value("abc") == "Invalid Input"
89 assert classify_value(0) == "Zero"
90 assert classify_value(-4) == "Even"
91
92 print("All classify_value tests passed successfully.")
93
```

```
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS_8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS_8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI_AC_assignments/ASS_8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function `is_palindrome(text)` and implement the function.

- Requirements:

- o Ignore case, spaces, and punctuation.

- o Handle edge cases such as empty strings and single characters.

Example Assert Test Cases:

```
assert is_palindrome("Madam") == True
```

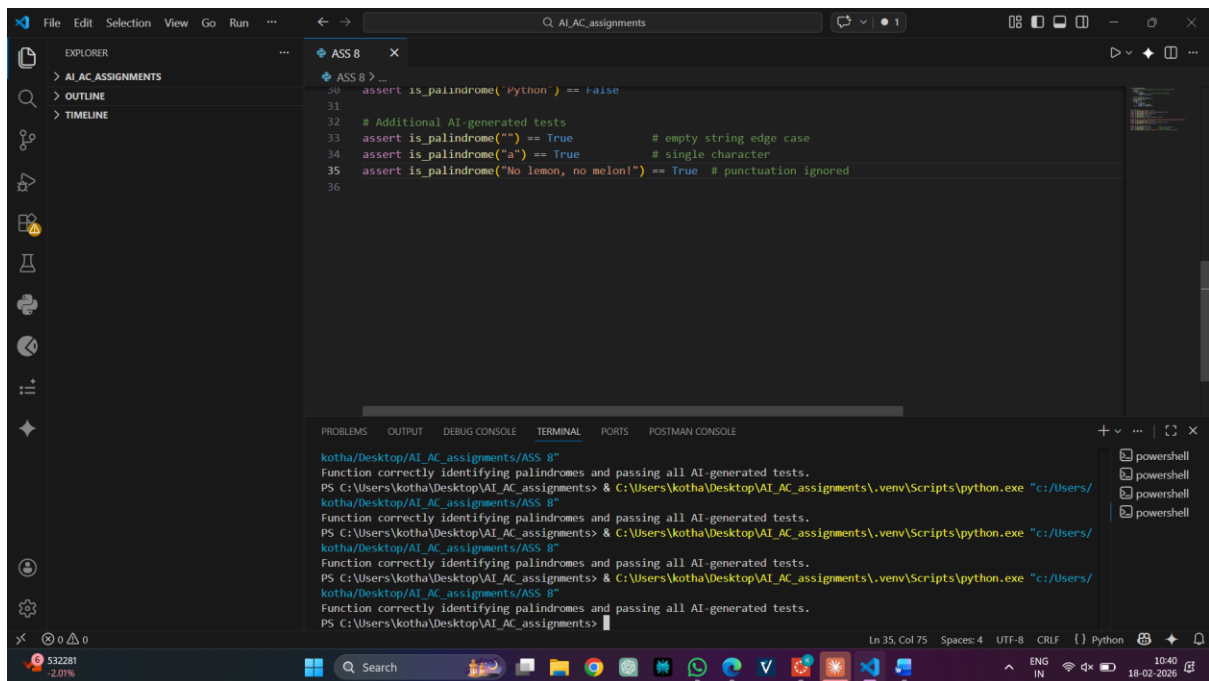
```
assert is_palindrome("A man a plan a canal Panama") ==
```

```
True
```

```
assert is_palindrome("Python") == False
```

Expected Output #3:

- Function correctly identifying palindromes and passing all AI-generated tests.



```
File Edit Selection View Go Run ... ALAC_assignments 1
EXPLORER
  > ALAC_ASSIGNMENTS
  > OUTLINE
  > TIMELINE
ASS 8
31 assert is_palindrome('Python') == False
32 # Additional AI-generated tests
33 assert is_palindrome("") == True # empty string edge case
34 assert is_palindrome("a") == True # single character
35 assert is_palindrome("No lemon, no melon!") == True # punctuation ignored
36

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
kotha/Desktop/ALAC_assignments/ASS 8
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/ALAC_assignments/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/ALAC_assignments/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI_AC_assignments> & C:\Users\kotha\Desktop\AI_AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/ALAC_assignments/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI_AC_assignments>
```

Task Description #4 (BankAccount Class – Apply AI for

Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases for a BankAccount class and then implement the class.

- Methods:

- o deposit(amount)

- o withdraw(amount)

- o get_balance()

Example Assert Test Cases:

```
acc = BankAccount(1000)
```

```
acc.deposit(500)
```

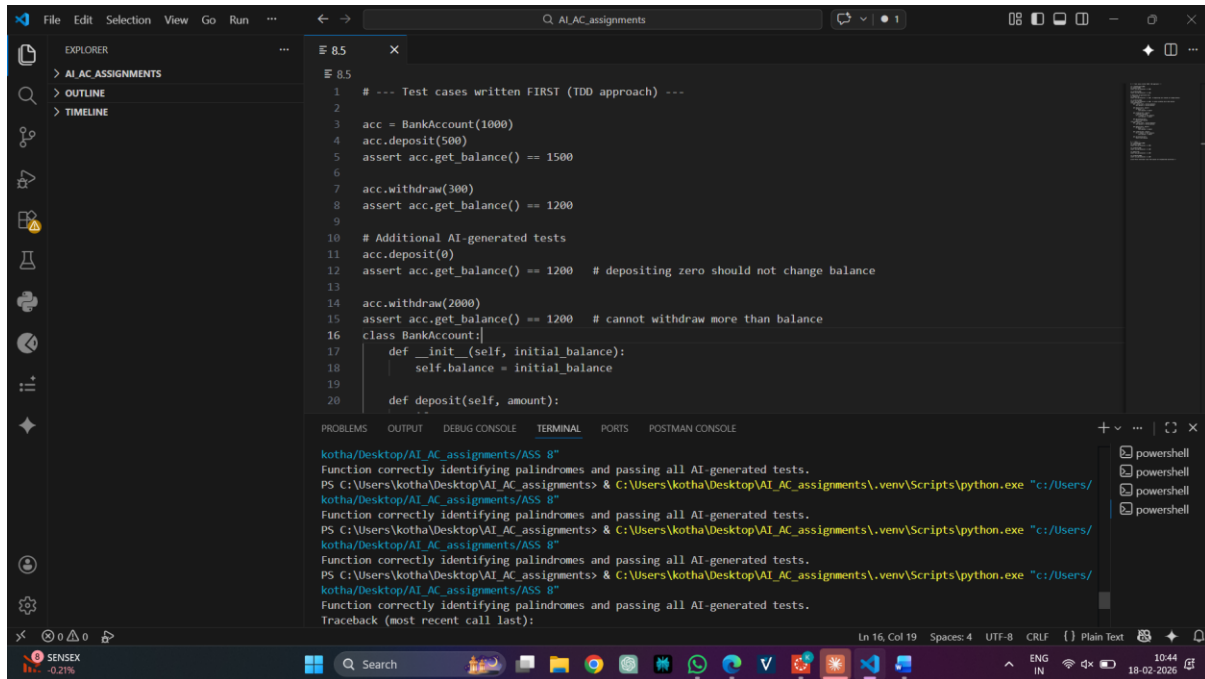
```
assert acc.get_balance() == 1500
```

```
acc.withdraw(300)
```

```
assert acc.get_balance() == 1200
```

Expected Output #4:

- Fully functional class that passes all AI-generated assertions.



```
8.5
1  # --- Test cases written FIRST (TDD approach) ---
2
3  acc = BankAccount(1000)
4  acc.deposit(500)
5  assert acc.get_balance() == 1500
6
7  acc.withdraw(300)
8  assert acc.get_balance() == 1200
9
10 # Additional AI-generated tests
11 acc.deposit(0)
12 assert acc.get_balance() == 1200 # depositing zero should not change balance
13
14 acc.withdraw(2000)
15 assert acc.get_balance() == 1200 # cannot withdraw more than balance
16 class BankAccount:
17     def __init__(self, initial_balance):
18         self.balance = initial_balance
19
20     def deposit(self, amount):
21
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```

```
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS> & C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS\venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI.AC_ASSIGNMENTS/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS> & C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS\venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI.AC_ASSIGNMENTS/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
PS C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS> & C:\Users\kotha\Desktop\AI.AC_ASSIGNMENTS\venv\Scripts\python.exe "c:/Users/kotha/Desktop/AI.AC_ASSIGNMENTS/ASS 8"
Function correctly identifying palindromes and passing all AI-generated tests.
Traceback (most recent call last):
```

Task Description #5 (Email ID Validation – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for a function `validate_email(email)` and implement the function.

- Requirements:

- o Must contain @ and .
- o Must not start or end with special characters.
- o Should handle invalid formats gracefully.

Example Assert Test Cases:

```
assert validate_email("user@example.com") == True
```

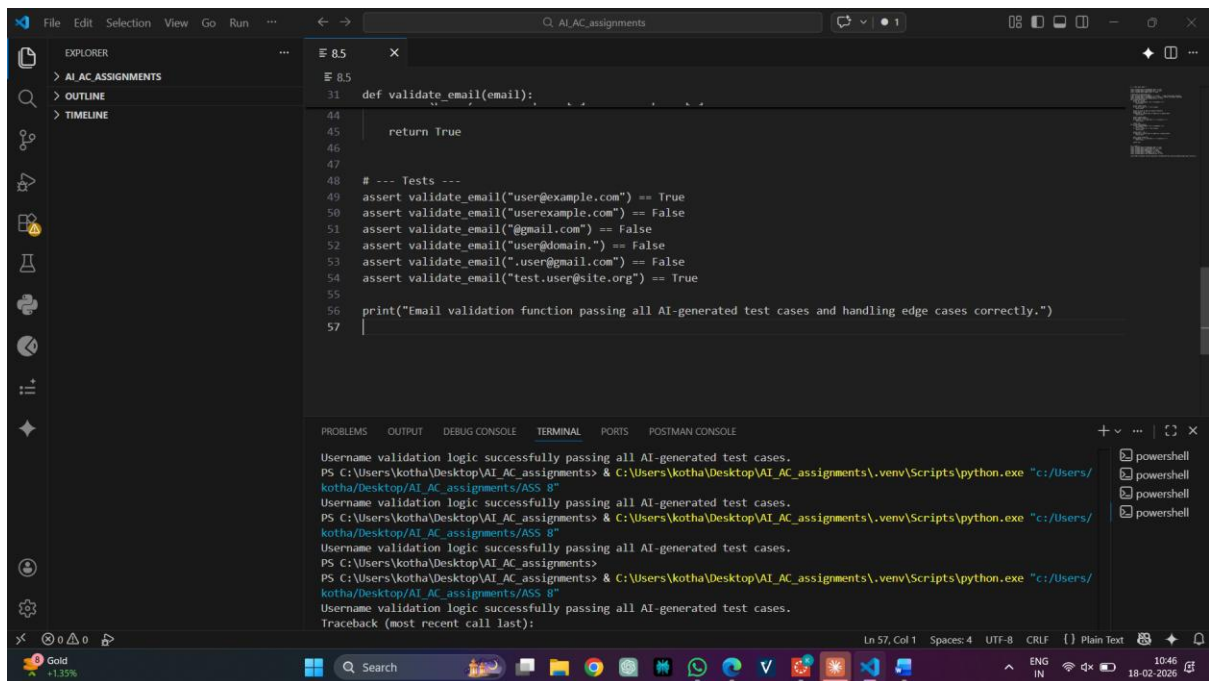
```
assert validate_email("userexample.com") == False
```

```
assert validate_email("@gmail.com") == False
```

Expected Output #5:

- Email validation function passing all AI-generated test cases

and handling edge cases correctly.



The image shows a Visual Studio Code editor window with a Python file named `AI.AC_assignments`. The code defines a `validate_email` function and includes several test cases using `assert` statements. The terminal at the bottom shows the command `python AI.AC_assignments` being executed, resulting in a `Traceback (most recent call last):` error. The error message indicates that the `validate_email` function is not defined, which is a contradiction to the code shown in the editor. The terminal output also includes the text "Username validation logic successfully passing all AI-generated test cases." repeated multiple times.

```
File Edit Selection View Go Run ... 8.5 x
> AI.AC_ASSIGNMENTS
> OUTLINE
> TIMELINE

31 def validate_email(email):
44     return True
45
46
47
48 # --- Tests ---
49 assert validate_email("user@example.com") == True
50 assert validate_email("userexample.com") == False
51 assert validate_email("@gmail.com") == False
52 assert validate_email("user@domain.") == False
53 assert validate_email(".user@gmail.com") == False
54 assert validate_email("test.user@site.org") == True
55
56 print("Email validation function passing all AI-generated test cases and handling edge cases correctly.")
57

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI.AC_assignments> & C:\Users\kotha\Desktop\AI.AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/AI.AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI.AC_assignments> & C:\Users\kotha\Desktop\AI.AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/AI.AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
PS C:\Users\kotha\Desktop\AI.AC_assignments>
PS C:\Users\kotha\Desktop\AI.AC_assignments> & C:\Users\kotha\Desktop\AI.AC_assignments\.venv\Scripts\python.exe "c:/Users/
kotha/Desktop/AI.AC_assignments/ASS 8"
Username validation logic successfully passing all AI-generated test cases.
Traceback (most recent call last):
  File "AI.AC_assignments", line 31, in <module>
    validate_email(email)
NameError: name 'validate_email' is not defined

Ln 57, Col 1 Spaces: 4 UTF-8 CRLF {} Plain Text 10:46 18-02-2026
```